Plastic Waste Mapping - Accesco Living

Grocery/FMCG Supply Chain & Dark-Store Operations

Research Objective

The purpose of this research is to identify types of plastic waste that may be produced along the grocery/FMCG supply chain and in a dark-store at Accesco Living. It examines plastic waste from the delivery, packing, transportation, storage, product packing and reverse logistics. The aim is to gain an understanding of the sources of plastic waste and potential re-use and recycle opportunities.

1. Plastic Waste at Supplier/Farm Stage

Plastic materials can be used in this stage for handling, protection, packaging and transport of grocery and FMCG products before their arrival in Accesco's facilities, at the supplier and farm stages. Plastic waste that can occur at this stage is plastic bags, plastic crates, plastic trays, plastic liners, protective wrapping and packaging for FMCG products. Damaged/rejected packaging could also turn into plastic waste. Certain materials, like crates and trays, might be used more than once and are best considered a reusable asset, not waste. Accesco's grocery business involves buying products from the farms and suppliers; this aspect is crucial in locating where plastic materials are getting into the supply chain.

2. Plastic Waste During Transportation

Plastic can be used in transit to protect products and ensure the security of goods in transit from suppliers or farms to Accesco's operational sites. Plastic waste that can be recycled may consist of stretch film, shrink wrap, plastic straps, protective plastic sheets, plastic liners, and damaged plastic crates. These materials can end up being wasted once they have been taken after transportation or if they are damaged and not usable for the next service. Recycling opportunities for clean and suitable plastic films and other items may exist, and reusable crates and containers could also be kept in use rather than discarded.

3. Plastic Waste at Dark-Store Operations

Dark stores are an important part of grocery fulfillment because products are received, stored, organized, and prepared for customer orders at these locations. Plastic waste may be generated when products arrive from suppliers, when packaging is removed, or when products are inspected and stored. Potential waste can include damaged bottles, torn pouches, plastic wrappers, stretch film, shrink wrap, plastic liners, damaged containers, and unusable packaging. Damaged or contaminated plastic may require separate handling, while reusable crates, totes, and containers should be separated from actual waste so they can continue to be used.

4. Plastic Waste During Picking and Packing

Product is picked and packaged from the dark store as per customer orders. Plastic materials can be used as a means of transportation, protection, assembly or securing of the products prior to delivery. Plastic waste can occur in this stage as plastic bags, delivery bags, plastic film, plastic liners, protective plastic, plastic tape and plastic labels. Packaging damage due to packing operations can also be wasted. The amount and variety of plastic used at this stage should be quantified to determine if there is a potential to reduce unnecessary single-use plastic through the use of reusable packaging or other materials.

5. Plastic Waste During Delivery

During the delivery stage, plastic may be present in both the products being delivered and the packaging used to transport customer orders. Potential plastic materials include delivery bags, reusable containers, plastic bottles, food containers, plastic cups, lids, wrappers, and other FMCG product packaging. Plastic packaging can become waste after customers use the products, while reusable delivery bags, crates, or containers may be collected and returned for further use. Damaged packaging during delivery should also be separated so that suitable materials can be recovered or recycled.

6. Customer Returns and Reverse Logistics

Some plastic materials may continue to be kept with the customer following order and could be recovered by a reverse-logistics system. These can be reusable delivery bags, containers, crates, totes and some recyclable plastic packaging. Materials can be returned to the delivery partners or designated collection points and delivered to an Accesco facility for sorting. The reusable items are cleaned, inspected, and returned to use and the recyclable plastic can be sorted by material type and placed in the correct recycling stream. Recycled plastic material that is contaminated or hard to recycle might need a different recovery/disposal method. This reverse collection can aid in the identification of plastics which may be missed from the supply chain as waste.

7. Plastic Waste Inventory

Plastic Type (Examples)	Source	Recovery
Rigid packaging: PET water bottle, HDPE detergent bottle, PP food container, sauce jar, lids/caps	FMCG	Recycling
Cups & food containers: drink cup, meal container	FMCG/Food	Reuse / recycling (if accepted)
Flexible packaging: milk pouch, biscuit wrapper	FMCG	Material-dependent
Sachets & multilayer: shampoo sachet, chips packet	FMCG	Difficult
Bags: grocery bag, delivery bag	Packing	Reuse / recycling
Transport film: stretch film, shrink wrap	Transport	Recycle (if clean/suitable)
Plastic straps: carton securing	Transport	Material-dependent
Crates & trays: produce crate, produce tray	Supplier	Reuse / repair / recycling
Protective materials: bubble wrap, liner, protective sheet	Supplier Chain	Reuse / recycling / material-dependent
Labels & tape	Packing	Limited
Damaged packaging: torn pouch/bottle, broken crate	Supply chain / Supplier	Material-dependent / repair / recycling

Key Finding

The research reveals that potential sources of plastic waste may occur throughout Accesco Living's grocery and FMCG activities: from farms and suppliers, during transport, in dark-stores, during picking and packing, on delivery and at the customers' home. Plastic products can range from bottles, containers, bags, pouches and wrappers, films, crates, trays and other packaging. Don't mix reusable items, including crates, trays, totes and delivery bags, with actual waste and allow them to be reused or repaired. Depending on the plastic's material, condition, contamination and recycling facilities, other plastics can be recyclable. Different recovery methods may be necessary for multilayer and contaminated plastics.

Information Accesco Should Collect

It is recommended to gather the data related to the type of plastic waste produced and quantities at each stage of operations to improve plastic waste management in Accesco. This may be the quantity of plastic waste generated by each dark store, the number of reusable containers or crates that are in use, reusable container or crate damage, end-of-life reusable container and crate, customer returned plastic containers, current plastic recycling operations, and the volume of plastic that is reused, recycled, or thrown away. This data can be used to guide Accesco in determining the primary sources of plastic waste, and in designing a successful reverse-commerce and recovery system.

References

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3. Central Pollution Control Board (CPCB) (n.d.) *Plastic Waste Management*. Available at: CPCB Plastic Waste Management Portal

4. Accesco Living (n.d.) *About Us*. Available at: Accesco Living – About Us